TP PROXMOX — DOCUMENTATION EXACTE DE SESSION

Commandes reelles et sorties exactes — Session du 9 juin 2026

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TP1 — Installation et configuration de Proxmox VE

Partie 1 — Preparation de l'hote Proxmox

Environnement materiel

Proxmox VE installe dans une VM VMware Workstation sur Windows. Verification du CPU hote depuis PowerShell :

```
PS C:\Windows\system32> Get-WmiObject Win32_Processor | Select-Object Name,
NumberOfCores, NumberOfLogicalProcessors

Name                                     NumberOfCores
NumberOfLogicalProcessors
----                                     ------------- ----------------------
-
13th Gen Intel(R) Core(TM) i7-13650HX            14
20
PS C:\Windows\system32>
```

Note : i7-13650HX : 6 P-cores (hyperthreades) + 8 E-cores = 14 cores / 20 threads. VM configuree avec 1 socket x 16 cores (bonne pratique vs 8 sockets x 1 core).

Configuration des depots Proxmox 9.x

Proxmox 9.x utilise Debian Trixie et le format DEB822 (.sources). Les depots Enterprise generaient des erreurs 401 :

```
root@proxmox:~# cat /etc/apt/sources.list.d/*.list
# desactive
# desactive
root@proxmox:~# ls /etc/apt/sources.list.d/
ceph.list  ceph.sources  debian.sources  pve-enterprise.list  pve-
enterprise.sources
```

Erreur : Les fichiers .sources Enterprise persistaient malgre la desactivation des .list — specifique a Proxmox 9.x sur Debian Trixie.

Solution : Vider les .sources Enterprise ET creer un nouveau fichier pve-no-subscription.sources avec la suite 'trixie'.

```
root@proxmox:~# echo "" > /etc/apt/sources.list.d/pve-enterprise.sources
root@proxmox:~# echo "" > /etc/apt/sources.list.d/ceph.sources
root@proxmox:~# cat > /etc/apt/sources.list.d/pve-no-subscription.sources <<
'EOF'
Types: deb
URIs: http://download.proxmox.com/debian/pve
Suites: trixie
Components: pve-no-subscription
EOF
root@proxmox:~# apt update
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://security.debian.org/debian-security trixie-security InRelease
Hit:4 http://download.proxmox.com/debian/pve bookworm InRelease
Get:5 http://download.proxmox.com/debian/pve trixie InRelease [3,534 B]
Get:6 http://download.proxmox.com/debian/pve trixie/pve-no-subscription
amd64 Packages [465 kB]
Fetched 469 kB in 0s (1,789 kB/s)
```

```
40 packages can be upgraded.
root@proxmox:~#
```

Telechargement ISO Debian

Erreur : debian-12.11.0-amd64-netinst.iso retournait 404 — Proxmox 9.x est base sur Debian 13, pas 12.

```
root@proxmox:/var/lib/vz/template/iso# curl -s
https://cdimage.debian.org/debian-cd/current/amd64/iso-cd/ | grep -o
'debian-[^"]*netinst.iso' | head -1
debian-13.5.0-amd64-netinst.iso
root@proxmox:/var/lib/vz/template/iso# wget
https://cdimage.debian.org/debian-cd/current/amd64/iso-cd/debian-13.5.0-
amd64-netinst.iso
debian-13.5.0-amd64-netinst.iso  100%[===============================>]
755.00M  24.9MB/s    in 33s
2026-06-09 11:46:28 (22.9 MB/s) - 'debian-13.5.0-amd64-netinst.iso' saved
[791674880/791674880]
root@proxmox:/var/lib/vz/template/iso#
```

Verification de l'hote Proxmox

```
root@proxmox:~# hostnamectl
  Static hostname: proxmox
    Icon name: computer-vm
      Chassis: vm
   Machine ID: 976fb44992164b07bf7a4a0a77cd60c1
Operating System: Debian GNU/Linux 13 (trixie)
          Kernel: Linux 7.0.2-6-pve
    Architecture: x86-64
 Virtualization: vmware
root@proxmox:~# cat /etc/resolv.conf
search lab.locall
nameserver 192.168.131.2
root@proxmox:~# ping -c 3 google.com
PING google.com (142.251.39.206) 56(84) bytes of data.
64 bytes from pnpara-ai-in-f14.1e100.net (142.251.39.206): icmp_seq=1
ttl=128 time=4.24 ms
64 bytes from pnpara-ai-in-f14.1e100.net (142.251.39.206): icmp_seq=2
ttl=128 time=4.07 ms
64 bytes from pnpara-ai-in-f14.1e100.net (142.251.39.206): icmp_seq=3
ttl=128 time=5.11 ms
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss
root@proxmox:~#
```

Parties 2, 3 et 4 — VM de base, QEMU Guest Agent, nettoyage

Installation de Debian 13 dans la VM 9000

VM creee avec : 1 socket x 2 cores, 2 Go RAM, 8 Go disque SCSI VirtIO, carte reseau VirtIO, machine q35, QEMU Agent active. Installation Debian avec uniquement SSH server + utilitaires systeme (pas d'interface graphique).

Note : L'identifiant 'admin' est reserve sur Debian 13. Utilisation de 'labuser' a la place.

Connexion SSH et passage root

```
C:\Users\Dimitri ANGELY>ssh labuser@192.168.131.138
labuser@192.168.131.138's password:
Linux web01 6.12.90+deb13.1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.90-2
(2026-05-27) x86_64
labuser@web01:~$ sudo -i
root@web01:~#
```

Note : 'su -' echouait car le mot de passe root avait ete laisse vide (bonne pratique template). 'sudo -i' utilise le mot de passe de labuser.

Installation QEMU Guest Agent et cloud-init

```
root@debian:~# apt update && apt upgrade -y
Tous les paquets sont a jour.
root@debian:~# apt install -y qemu-guest-agent cloud-init
root@debian:~# systemctl enable qemu-guest-agent
Synchronizing state of qemu-guest-agent.service...
The unit files have no installation config (WantedBy=, RequiredBy=...)
(This means they are not meant to be enabled or disabled using systemctl.)
root@debian:~# systemctl start qemu-guest-agent
root@debian:~#
```

Note : Le message 'no installation config' est normal pour qemu-guest-agent — il s'active via udev quand le canal QEMU est disponible, pas via systemctl enable.

Nettoyage avant conversion en template

```
root@debian:~# apt autoremove -y && apt clean
root@debian:~# cat /dev/null > ~/.bash_history
root@debian:~# truncate -s 0 /etc/machine-id
root@debian:~# rm -f /var/lib/dbus/machine-id
root@debian:~# ln -s /etc/machine-id /var/lib/dbus/machine-id
root@debian:~# rm -f /etc/ssh/ssh_host_*
root@debian:~# shutdown -h now
```

Explication de chaque etape :

- machine-id vide : evite que tous les clones aient le meme identifiant reseau
- cles SSH supprimees : chaque clone regenere ses propres cles au premier boot
- historique bash vide : ne pas transmettre d'informations sensibles aux clones

Configuration Cloud-Init et conversion en template

Depuis Proxmox : Matériel > Ajouter > Lecteur CloudInit (local-lvm). Onglet Cloud-Init configure avec ip=dhcp. Puis clic droit sur la VM > Convertir en modele.

Partie 6 — Creation des clones

VM ID	Nom	Type	Role
101	web01	Clone complet	Serveur web
102	db01	Clone lie	Serveur base de donnees
103	admin01	Clone lie	Administration
104	test-rollback01	Clone lie	Test snapshot/rollback

Apres demarrage, les 4 VMs obtiennent des IPs distinctes via DHCP (cloud-init) :

```
web01           192.168.131.138
db01            192.168.131.139
admin01         192.168.131.140
test-rollback01 192.168.131.141
```

Partie 7 — Personnalisation des VMs

web01 — Installation Apache

```
root@web01:~# apt install -y apache2
root@web01:~# echo '<!DOCTYPE html><meta charset="UTF-8"><h1>Serveur web01
operationnel</h1>' > /var/www/html/index.html
root@web01:~# systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled)
     Active: active (running) since Tue 2026-06-09 13:16:40 CEST; 37s ago
   Main PID: 1510 (apache2)
root@web01:~#
```

Test depuis Windows :

```
C:\Users\Dimitri ANGELY>curl http://192.168.131.138
<!DOCTYPE html><meta charset="UTF-8"><h1>Serveur web01 operationnel</h1>
C:\Users\Dimitri ANGELY>
```

db01 — Installation MariaDB

```
root@db01:~# apt install -y mariadb-server
root@db01:~# systemctl status mariadb
● mariadb.service - MariaDB 11.8.6 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled)
     Active: active (running) since Tue 2026-06-09 13:23:27 CEST; 21s ago
     Status: "Taking your SQL requests now..."
   Main PID: 1811 (mariadbd)
root@db01:~#
```

admin01 — Tests reseau

```
root@admin01:~# apt install -y curl nmap
root@admin01:~# curl http://192.168.131.138
```

```
<!DOCTYPE html><meta charset="UTF-8"><h1>Serveur web01 operationnel</h1>
root@admin01:~# ping -c 3 192.168.131.139
PING 192.168.131.139 (192.168.131.139) 56(84) bytes of data.
64 bytes from 192.168.131.139: icmp_seq=1 ttl=64 time=0.747 ms
64 bytes from 192.168.131.139: icmp_seq=2 ttl=64 time=1.15 ms
3 packets transmitted, 3 received, 0% packet loss
root@admin01:~# nmap -p 3306 192.168.131.139
PORT     STATE  SERVICE
3306/tcp closed mysql
root@admin01:~#
```

Note : Port 3306 'closed' est normal : MariaDB ecoute sur localhost uniquement par securite. 'Closed' = hote repond mais refuse, different de 'filtered' (pare-feu bloque).

Partie 8 — Snapshot, panne volontaire et rollback

1. Etat initial — Apache fonctionnel sur test-rollback01

```
root@test-rollback01:~# apt install -y apache2
root@test-rollback01:~# echo '<!DOCTYPE html><meta charset="UTF-8"><h1>test-
rollback01 operationnel</h1>' > /var/www/html/index.html
C:\Users\Dimitri ANGELY>curl http://192.168.131.141
<!DOCTYPE html><meta charset="UTF-8"><h1>test-rollback01 operationnel</h1>
```

2. Creation du snapshot 'avant-panne'

Depuis Proxmox : VM 104 > Instantanes > Creer un instantane

- Nom : avant-panne
- Description : Apache fonctionnel
- Inclure la RAM : decoche

```
Logical volume "vm-104-state-avant-panne" created.
Logical volume pve/vm-104-state-avant-panne changed.
saving VM state and RAM using storage 'local-lvm'
298.00 B in 0s
```

3. Panne provoquee

```
root@test-rollback01:~# systemctl stop apache2
root@test-rollback01:~# systemctl disable apache2
root@test-rollback01:~# rm /var/www/html/index.html
```

Constat de la panne depuis Windows :

```
C:\Users\Dimitri ANGELY>curl http://192.168.131.141
curl: (7) Failed to connect to 192.168.131.141 port 80 after 2054 ms: Could
not connect to server
C:\Users\Dimitri ANGELY>
```

4. Rollback et verification

Depuis Proxmox : VM 104 > Instantanes > selectionner 'avant-panne' > Retour en arriere > Oui

```
C:\Users\Dimitri ANGELY>curl http://192.168.131.141
<!DOCTYPE html><meta charset="UTF-8"><h1>test-rollback01 operationnel</h1>
C:\Users\Dimitri ANGELY>
```

Solution : Le rollback restaure exactement l'etat du snapshot. Apache repond avec la page originale.

TP2 — Architecture reseau segmentee avec VLANs

Partie 1 — Creation des reseaux Proxmox

Depuis Proxmox : proxmox > Reseau > Creer > Bridge Linux

- Nom : vmbr1
- Gere les VLAN : coche (VLAN-aware)
- Bridge ports : vide (bridge interne)

VLAN	Zone	Reseau	Gateway fw01
10	MGMT	192.168.10.0/24	192.168.10.254
20	SRV	192.168.20.0/24	192.168.20.254
30	USERS	192.168.30.0/24	192.168.30.254
40	DMZ	192.168.40.0/24	192.168.40.254

Partie 2 — Deploiement et configuration de fw01

Interfaces reseau de fw01 (5 interfaces)

Interface	Bridge	VLAN	Zone	IP
eth0	vmbr0	-	Management Proxmox	192.168.131.142 (DHCP)
ens19	vmbr1	20	SRV	192.168.20.254
ens20	vmbr1	30	USERS	192.168.30.254
ens21	vmbr1	40	DMZ	192.168.40.254
ens22	vmbr1	10	MGMT	192.168.10.254

Note : ens22 a ete ajoutee a chaud (hot-plug) apres constat que le VLAN 10 MGMT manquait — Proxmox supporte le branchement a chaud des cartes VirtIO.

Verification des interfaces fw01

```
root@fw01:~# ip a
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP>
    inet 192.168.131.142/24 brd 192.168.131.255 scope global dynamic
noprefixroute eth0
3: ens19: <BROADCAST,MULTICAST,UP,LOWER_UP>
    inet 192.168.20.254/24 brd 192.168.20.255 scope global ens19
4: ens20: <BROADCAST,MULTICAST,UP,LOWER_UP>
    inet 192.168.30.254/24 brd 192.168.30.255 scope global ens20
5: ens21: <BROADCAST,MULTICAST,UP,LOWER_UP>
    inet 192.168.40.254/24 brd 192.168.40.255 scope global ens21
6: ens22: <BROADCAST,MULTICAST,UP,LOWER_UP>
    inet 192.168.10.254/24 brd 192.168.10.255 scope global ens22
root@fw01:~#
```

Activation du routage IP

```
root@fw01:~# echo 1 > /proc/sys/net/ipv4/ip_forward
root@fw01:~# echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf
root@fw01:~# sysctl net.ipv4.ip_forward
net.ipv4.ip_forward = 1
root@fw01:~#
```

Configuration nftables (NAT + filtrage)

```
root@fw01:~# nft list ruleset
table inet filter {
        chain input {
                type filter hook input priority filter; policy accept;
        }
        chain forward {
                type filter hook forward priority filter; policy accept;
                iifname "ens20" oifname "ens19" drop
                iifname "ens19" oifname "ens20" drop
        }
        chain output {
                type filter hook output priority filter; policy accept;
        }
}
table ip nat {
        chain postrouting {
                type nat hook postrouting priority srcnat; policy accept;
                oifname "eth0" masquerade
        }
}
root@fw01:~#
```

Partie 3 — Configuration des VMs par zone

web-dmz01 (Zone DMZ — 192.168.40.10)

```
root@web-dmz01:~# ip a show ens19
3: ens19: <BROADCAST,MULTICAST,UP,LOWER_UP>
    inet 192.168.40.10/24 brd 192.168.40.255 scope global ens19
root@web-dmz01:~# systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
     Active: active (running) since Tue 2026-06-09 15:19:06 CEST; 32s ago
   Main PID: 1674 (apache2)
root@web-dmz01:~#
```

db-srv01 (Zone SRV — 192.168.20.10)

```
root@db-srv01:~# ip a show ens19
3: ens19: <BROADCAST,MULTICAST,UP,LOWER_UP>
    inet 192.168.20.10/24 brd 192.168.20.255 scope global ens19
root@db-srv01:~# systemctl status mariadb --no-pager
● mariadb.service - MariaDB 11.8.6 database server
     Active: active (running) since Tue 2026-06-09 13:23:27 CEST
     Status: "Taking your SQL requests now..."
root@db-srv01:~#
```

client01 (Zone USERS — 192.168.30.10)

```
root@client01:~# ip a show ens19
3: ens19: <BROADCAST,MULTICAST,UP,LOWER_UP>
    inet 192.168.30.10/24 brd 192.168.30.255 scope global ens19
root@client01:~#
```

admin-mgmt01 (Zone MGMT — 192.168.10.10)

```
root@admin-mgmt01:~# ip a show ens19
3: ens19: <BROADCAST,MULTICAST,UP,LOWER_UP>
    inet 192.168.10.10/24 brd 192.168.10.255 scope global ens19
root@admin-mgmt01:~#
```

Partie 4 — Tests de connectivite et filtrage

Tests depuis admin-mgmt01

```
root@admin-mgmt01:~# ping -c 3 192.168.10.254
PING 192.168.10.254 (192.168.10.254) 56(84) bytes of data.
64 bytes from 192.168.10.254: icmp_seq=1 ttl=64 time=2.00 ms
64 bytes from 192.168.10.254: icmp_seq=2 ttl=64 time=0.924 ms
64 bytes from 192.168.10.254: icmp_seq=3 ttl=64 time=1.45 ms
3 packets transmitted, 3 received, 0% packet loss

root@admin-mgmt01:~# ping -c 3 192.168.40.10
PING 192.168.40.10 (192.168.40.10) 56(84) bytes of data.
64 bytes from 192.168.40.10: icmp_seq=1 ttl=63 time=3.56 ms
64 bytes from 192.168.40.10: icmp_seq=2 ttl=63 time=1.68 ms
64 bytes from 192.168.40.10: icmp_seq=3 ttl=63 time=1.33 ms
3 packets transmitted, 3 received, 0% packet loss

root@admin-mgmt01:~# curl http://192.168.40.10
<!DOCTYPE html><meta charset="UTF-8"><h1>Serveur web-dmz01 - Zone DMZ</h1>
root@admin-mgmt01:~#
```

Note : TTL 63 sur le ping vers 192.168.40.10 confirme le routage via fw01 — chaque routeur decremente le TTL de 1.

Tests depuis client01 — Filtrage nftables

```
root@client01:~# ping -c 3 192.168.30.254
64 bytes from 192.168.30.254: icmp_seq=1 ttl=64 time=1.05 ms
64 bytes from 192.168.30.254: icmp_seq=2 ttl=64 time=1.64 ms
3 packets transmitted, 3 received, 0% packet loss

root@client01:~# ping -c 3 192.168.40.10
64 bytes from 192.168.40.10: icmp_seq=1 ttl=63 time=1.29 ms
64 bytes from 192.168.40.10: icmp_seq=2 ttl=63 time=1.66 ms
3 packets transmitted, 3 received, 0% packet loss

root@client01:~# ping -c 3 192.168.20.10
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
--- 192.168.20.10 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2038ms
root@client01:~#
```

Source	Destination	Resultat	Explication
admin-mgmt01	192.168.10.254 (fw01 MGMT)	OK TTL 64	Meme zone, pas de routage
admin-mgmt01	192.168.40.10 (web-dmz01)	OK TTL 63	Route via fw01
admin-mgmt01	http://192.168.40.10	OK page web	Apache accessible
client01	192.168.30.254 (fw01 USERS)	OK TTL 64	Meme zone
client01	192.168.40.10 (web-dmz01)	OK TTL 63	Route via fw01

client01	192.168.20.10 (db-srv01)	BLOQUE 100% perte	Regle nftables drop USERS->SRV

Bilan final — Etat de l'infrastructure

Element	Detail	Statut
Proxmox VE	Version 9.2.3 sur Debian 13 (Trixie)	OK
Interface web	https://192.168.131.135:8006	OK
Depots	pve-no-subscription trixie actif	OK
ISO Debian	debian-13.5.0-amd64-netinst.iso (755 Mo)	OK
Template	VM 9000 tmpl-debian-13-base avec cloud-init	OK
QEMU Guest Agent	IP visible dans Proxmox pour toutes les VMs	OK
web01 (101)	Apache — 192.168.131.138	OK
db01 (102)	MariaDB 11.8.6 — 192.168.131.139	OK
admin01 (103)	Tests reseau — 192.168.131.140	OK
test-rollback01 (104)	Snapshot + panne + rollback valides	OK
vmbr1	Bridge VLAN-aware (VLANs 10/20/30/40)	OK
fw01 (200)	5 interfaces, ip_forward=1, NAT, nftables	OK
web-dmz01 (201)	Apache DMZ — 192.168.40.10	OK
db-srv01 (202)	MariaDB SRV — 192.168.20.10	OK
client01 (203)	USERS — 192.168.30.10	OK
admin-mgmt01 (204)	MGMT — 192.168.10.10	OK
Routage inter-VLAN	TTL 63 confirme le passage par fw01	OK
Filtrage USERS->SRV	100% packet loss — regle nftables drop	OK